

HEARDS - Trading Strategies

By Arshia Feizmohammady

Documentation

Link to official documentation for HEARDS Dataset:

<https://www.spglobal.com/content/dam/spglobal/ci/en/documents/platts/en/our-methodology/methodology-specifications/refined-products/refined-products-asia-pacific-middle-east-specifications.pdf>

Action Types

- Offer
 - ~16% of dataset
 - Firm intent to *sell*, with price and specifications.
- Bid
 - ~15% of dataset
 - Firm intent to *buy* a product, specifying price, volume, and delivery window.
- Raise
 - ~39% of dataset
 - Editorial note indicating upward price revision (not an actual trade, bid, or offer).
- Lower
 - ~30% of dataset
 - Editorial note indicating downward price revision.

How we will treat the action types

- **Raise / Lower** will be treated as **sentiment signals**, not as trades
 - We'll compute a daily **sentiment score** based on relative counts
 - We'll flag **high-activity days** (≥ 20 signals) as **event regimes**.
- **Bid / Offer** will be retained as **market intent** data, but not assumed to be completed trades.

I focused my analysis on MOPS 380, which makes up over 61% of the dataset and represents the most liquid and consistently reported benchmark. This choice allows me to maximize signal reliability while avoiding noise from the thinner MOPS 180 market.

How I Backtest Each Strategy

- For each trading day t , strategy signals will be generated using only data available up to $t-1$.
- Trades will be executed at **MOPS 380 benchmark close on day t** .
- No lookahead or overlap with unconfirmed OTC data.
- Daily sentiment and OTC indicators will be merged with historical MOPS 380 benchmark prices for simulation.

Metrics Used to Evaluate Strategies

- Total Return / PnL
 - Raw performance over the backtest window
- Sharpe Ratio
 - Measures risk-adjusted return
- Max Drawdown
 - Largest observed loss from peak
- Volatility
 - Standard deviation of daily returns
- Win Rate
 - Percentage of profitable trades
- Turnover
 - Measures trading activity / realism
- Average Holding Period
 - Helps compare tactical vs. persistent strategies

Backtesting Assumptions

Capital & Execution Assumptions

- **Starting Capital:** \$1 000 000 USD
- **Position Type:** Fully allocated (100% long, 100% short, or 0% cash)
- **Execution Price:** MOPS 380 benchmark **close on day t**
- **Trading Frequency:** Daily — signals recalculated each day
- **Holding Period:** Flexible — defined by each strategy logic
- **No Transaction Costs:** (Can be added later in sensitivity tests)

Risk Controls

- **No Leverage:** Position size limited to capital available
- **No Borrowing or Margin:** Cash cannot go below \$0
- **Bankruptcy Rule:** Simulation stops if capital falls to or below \$0

Accounting Methods

- $\text{Daily PnL} = (\text{Position return on day } t) \times (\text{Capital on day } t-1)$
- Capital is updated each day based on that day's PnL
- Strategies will track **cumulative capital**, **daily returns**, and **position flags** over time

Baseline Strategy Models

Buy & Hold

- Always fully long MOPS 380 from start to end of the backtest.
- Represents passive market exposure.
- PnL = price appreciation only.

Price Momentum

- **Signal:** 5-day return (return_{t-1_5d}) $> 0 \rightarrow$ enter **long**, else **cash**.
- Captures short-term trend-following behavior.

Volatility Targeting

- **Signal:** Scale position daily to target a fixed annualized volatility (e.g., 10%).
- **Position Sizing:** $\text{Position}_t = \text{Target Vol} / \text{Rolling Realized Vol}_{t-1}$, where realized volatility is estimated using a 20-day rolling standard deviation.
- **Rationale:** Maintains consistent risk exposure and avoids excessive leverage during high-volatility periods.

Buy & Hold

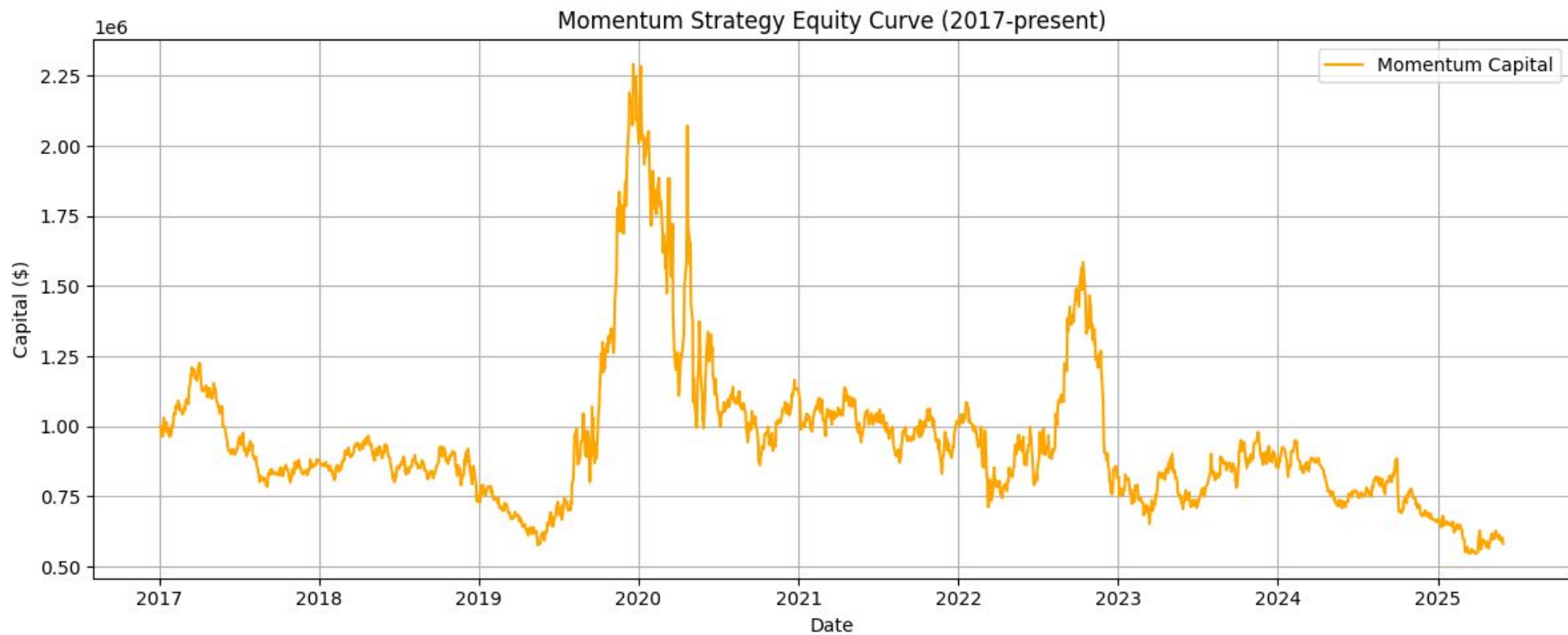


Buy & Hold

- Total Return: 26.43%
- Sharpe Ratio: 0.29
- Max Drawdown: -80.47%
- Annual Volatility: 45.19%
- Win Rate: 52.95%
- Turnover: 0
- Avg Hold Period: NA
- # of Trades: 1
- CAGR: 2.85%

Over an 8.4-year period, the Buy & Hold strategy delivered only modest compounded returns despite high volatility and large drawdowns — further motivating the need for timing-based or sentiment-enhanced approaches.

Price Momentum

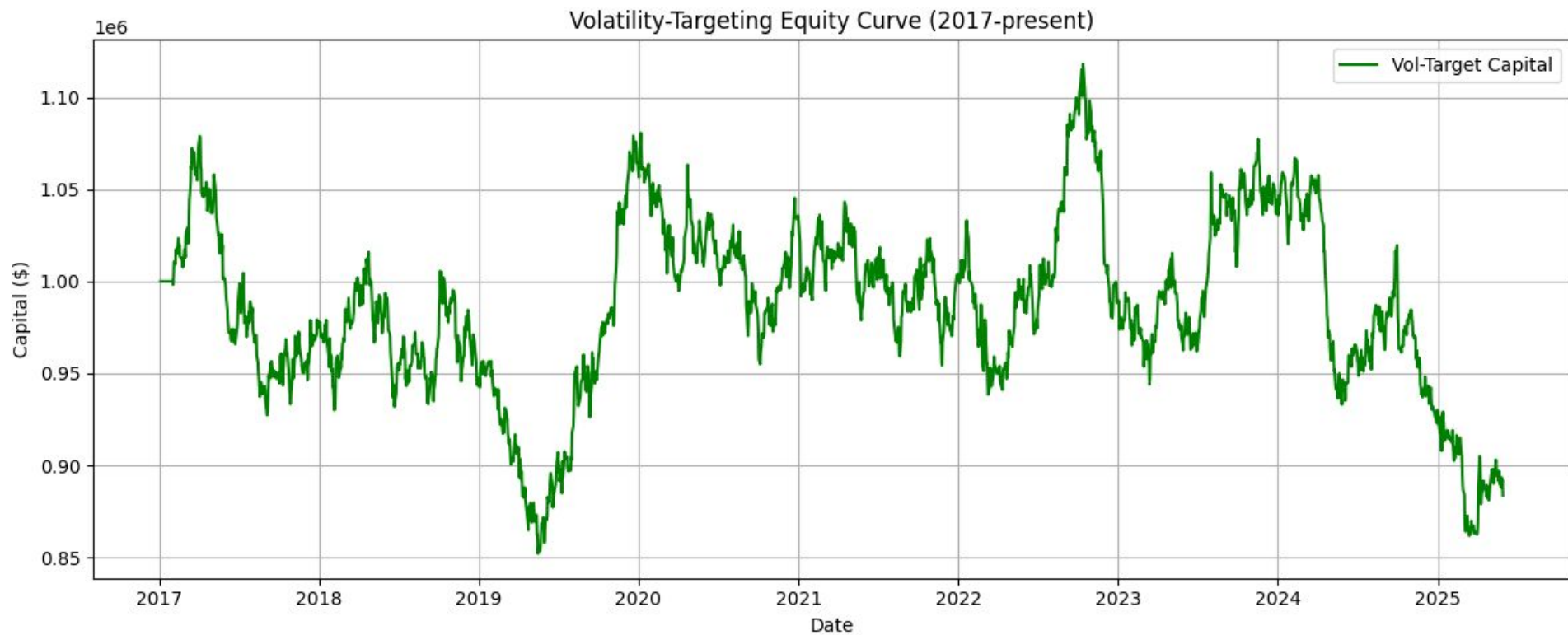


Price Momentum

- Total Return: -41.87%
- Sharpe Ratio: 0.08
- Max Drawdown: -76.16%
- Annual Volatility: 45.20%
- Win Rate: 48.79%
- Turnover: 1.02
- Avg Hold Period: 1.96
- # of Trades: 1076
- CAGR: -6.25

The Price Momentum strategy, while active with 1,076 trades and a turnover of 1.02, delivered a poor performance with a -41.87% return and a Sharpe ratio of 0.08. Its high annual volatility of 45.20% and max drawdown of -76.16% indicate extreme risk with little reward, despite a nearly even win rate of 48.79%.

Volatility Targeting



Volatility Targeting

- Total Return: -11.64%
- Sharpe Ratio: -0.10
- Max Drawdown: -22.91%
- Annual Volatility: 10.11%
- Win Rate: 48.41%
- Turnover: 0.3324
- Avg Hold Period: 1.01 days
- # of Trades: 2085
- CAGR: -1.46%

The Volatility Targeting strategy demonstrated much better risk control, with a lower annual volatility of 10.11% and a smaller drawdown of -22.91%. While still yielding a negative CAGR of -1.46%, its 2,085 trades and low turnover (0.33) show that the strategy maintained smoother capital allocation and tighter position sizing.

Strategy α -1 – Sentiment + Price Confirmation

- Signal (evaluated daily)
 - **Long +1:** $\text{directional_score} > \theta$ **and** yesterday's return > 0
 - **Short -1:** $\text{directional_score} < -\theta$ **and** yesterday's return < 0
 - Else flat 0
- **Exit:** close after k days **or** when score crosses 0
- **Sizing / Risk:** fully allocated; position scaled by 10 % target-vol overlay (same as baselines)
- **Parameter grid:** $\theta \in \{0.5, 1.0, 1.5\}$ $k \in \{3, 5, 7 \text{ days}\}$
- **Key fields used:** directional_score , OTC price
- **Objective:** beat all baselines on Sharpe & drawdown with turnover ≤ 1.2
- (Back-testing capital, execution price, and no-leverage rules identical to baseline assumptions.)

Sentiment + Price Confirmation



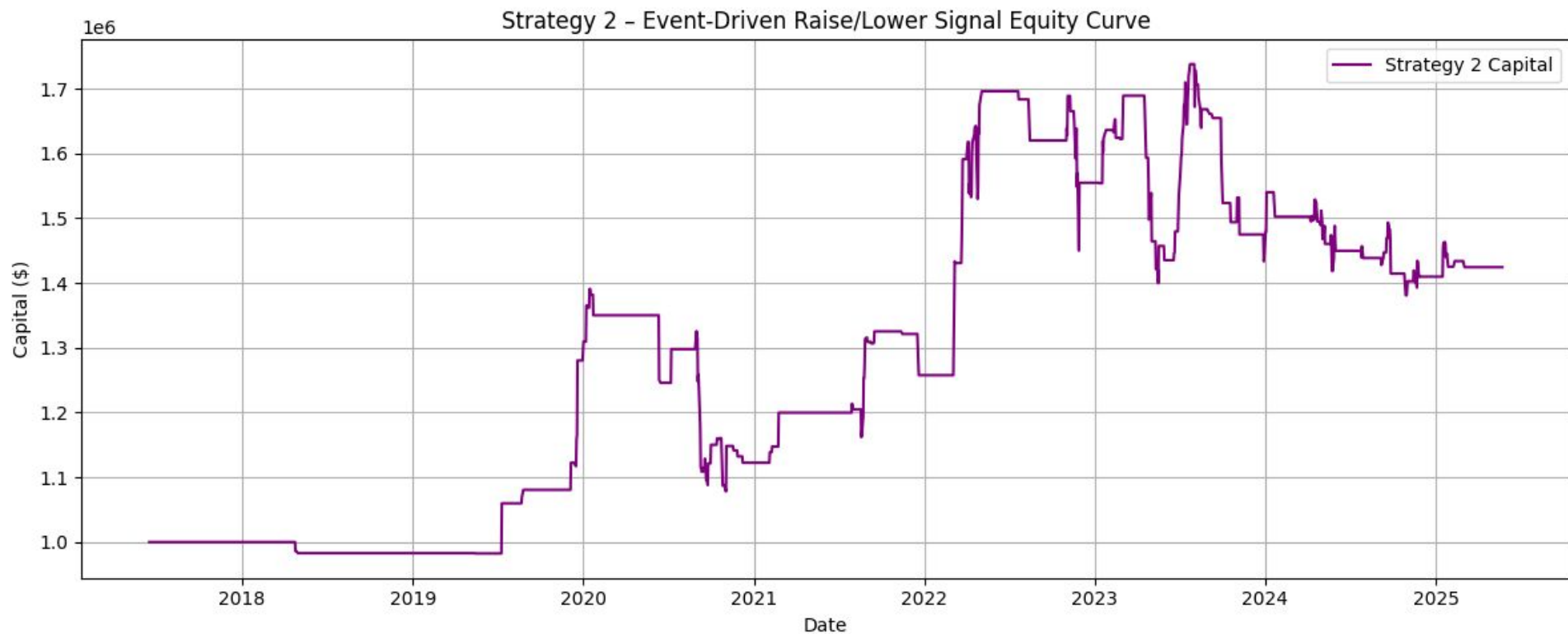
Sentiment + Price Confirmation

- Total Return: -33.13%
- Sharpe Ratio: -0.08
- Max Drawdown: -60.78%
- Annual Volatility: 25.10%
- Win Rate: 15.46%
- Turnover: 0.14
- Avg Hold Period: 7.36 days
- # of Trades: 270
- CAGR: -4.95

Strategy α -2 – Event-Driven Raise / Lower

- **Signal (scanned intraday headlines)**
 - **Short -1:** first **Lower** headline in past 24 h
 - **Long +1:** first **Raise / Offer** headline in past 24 h
 - No opposing event $\rightarrow$ retain position; simultaneous Raise & Lower $\Rightarrow$ flat 0
- **Exit:** auto-close after k days **or** on opposite event
- **Sizing / Risk:** position = min(headline-count last 24 h , 1) $\times$ sign; scaled by 10 % target-vol overlay
- **Parameter grid:** $k \in \{2, 4, 6 \text{ days}\}$ look-back window $m = 24 \text{ h}$ (sensitivity test $\pm 12 \text{ h}$)
- **Key fields used:** action_type (“Lower”, “Raise”, “Offer”), updatedAt timestamps
- **Objective:** capture discrete bearish/bullish price shocks identified in correlation study, with tighter drawdown than Momentum baseline and Sharpe $\geq$ Buy & Hold
- (Back-testing capital, execution price, and no-leverage rules identical to baseline assumptions.)

Event-Driven Raise / Lower



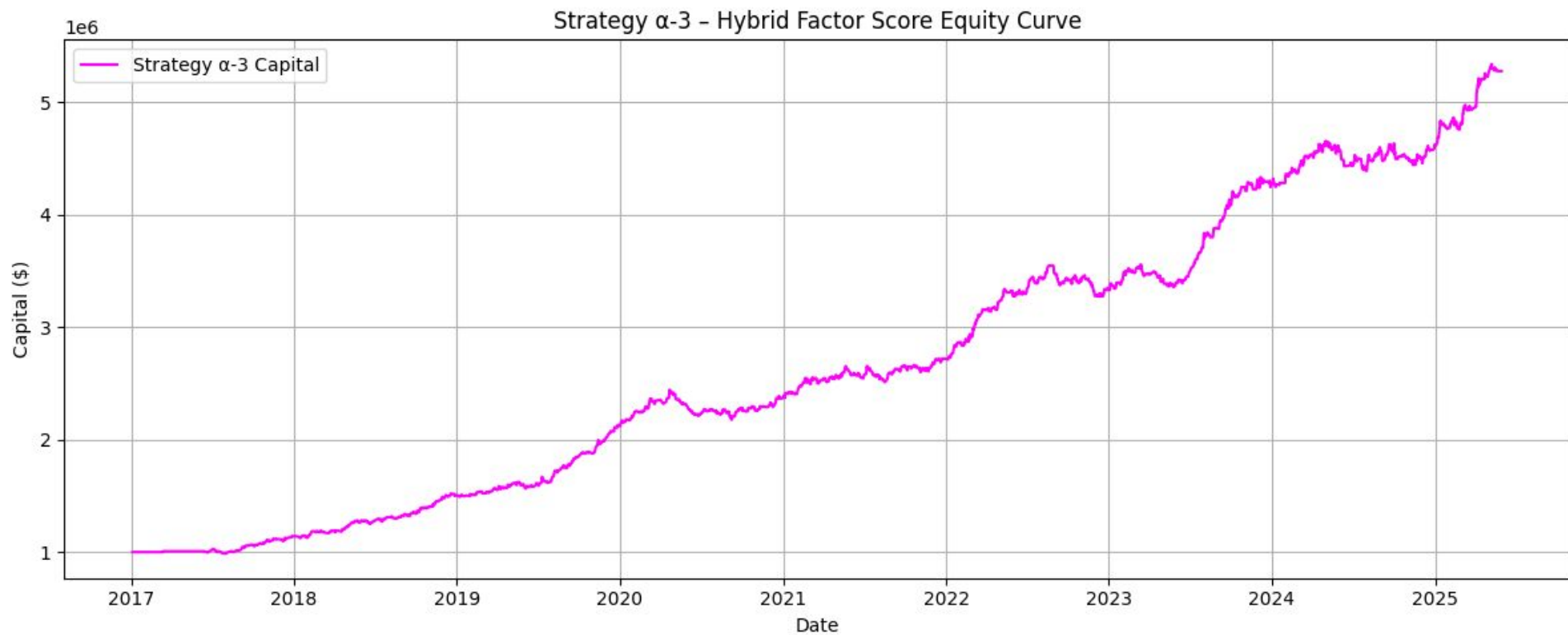
Event-Driven Raise / Lower

- Total Return: 42.43%
- Sharpe Ratio: 0.39
- Max Drawdown: -22.47%
- Annual Volatility: 14.16%
- Win Rate: 5.89%
- Turnover: 0.09
- Avg Hold Period: 10.74 days
- # of Trades: 185
- CAGR: 4.56%

Strategy α -3 – Hybrid Factor Score

- **Composite Score (computed each day)**
Score = $0.4 \times \text{z-Momentum}_{5\text{-day}} + 0.3 \times \text{Net Sentiment} + 0.3 \times \text{Event Flag}$
 - *z-Momentum*: z-score of 5-day OTC return
 - *Event Flag*: +1 for Raise/Offer, -1 for Lower, 0 otherwise
- **Position Rule** (sign + filter)
 - **Long +1** if Score > $+\lambda$ | **Short -1** if Score < $-\lambda$ | Flat otherwise
 - λ (signal threshold) prevents noise trades
- **Exit & Risk Management**
 - Hold until Score crosses 0 **or** max k days ($k \in \{5, 10\}$)
 - Position scaled by 10 % target-vol overlay (same as baselines)
- **Parameter grid**: $\lambda \in \{0.3, 0.5, 0.7\}$ $k \in \{5, 10\}$ momentum look-back $\in \{5, 10 \text{ days}\}$
- **CSV fields used**: price, net_sentiment, directional_score, action_type, updatedDate
- **Objective**: combine three highest-correlation drivers to surpass **all** baselines on Sharpe and keep max drawdown < 30

Hybrid Factor Score



Hybrid Factor Score

- Total Return: 427.77%
- Sharpe Ratio: 2.39
- Max Drawdown: -10.83%
- Annual Volatility: 8.48%
- Win Rate: 39.40%
- Turnover: 0.46
- Avg Hold Period: 1.75 days
- # of Trades: 1922
- CAGR: 21.90%

Strategy α -4 – ML Probabilistic Classifier

- **Model** Gradient-Boosted Trees (XGBoost) predicting **next-day up / down**
 - Features:
 - 20-day lags of OTC returns & volatility
 - directional_score, net_sentiment
 - Event dummies (Raise, Lower, Offer, Bid)
 - Rolling counts of each event last 3 d & 5 d
 - party, location, volume_min/max one-hot encoded
 - **Walk-forward retrain** every 6 months (60 % train 20 % val 20 % test)
- **Trading Rule**
 - **Long +1** if $P(\text{up}) > 0.60$ | **Short -1** if $P(\text{up}) < 0.40$ | Flat otherwise
 - Position size scaled by model confidence: $|P(\text{up}) - 0.5| \times \text{sign} \cdot \text{vol-cap} (10 \%)$
- **Risk & Execution** same capital / no-leverage framework as baselines
- **Key Hyper-params (grid)**
 - n-trees {200, 400} • depth {3, 5} • learning-rate {0.05, 0.10}
 - Probability threshold {0.55, 0.60, 0.65}
- **Objective** Demonstrate that a data-rich ML signal can beat all rule-based baselines on Sharpe **and** cut drawdown < 20 % while keeping turnover < 1.5.

ML Probabilistic Classifier



ML Probabilistic Classifier

- Total Return: -16.77%
- Sharpe Ratio: -0.27
- Max Drawdown: -30.11%
- Annual Volatility: 7.71%
- Win Rate: 39.66%
- Turnover: 0.19
- Avg Hold Period: 1.05 days
- # of Trades: 1824
- CAGR: -2.36%

Strategy α -5 – OTC $\leftrightarrow$ Benchmark Spread Mean-Reversion

- **Spread Definition**

$$\text{Spread}_t = \text{OTC Price}_t - \text{MOPS 380 Benchmark}_t$$

Z-score calculated on a 20-day rolling window.

- **Trade Rule Long OTC / Short Bench** when $Z < -z_0$

- **Short OTC / Long Bench** when $Z > +z_0$
- Exit when $|Z| < 0.3$ or after k days.

- **Position Sizing**

- Dollar-neutral pair; notional scaled by 10 % target-vol overlay (same risk cap as baselines).

- **Parameter Grid**

- $z_0 \in \{1.0, 1.5, 2.0\}$ $k \in \{5, 10\}$ look-back $\in \{20, 30 \text{ days}\}$

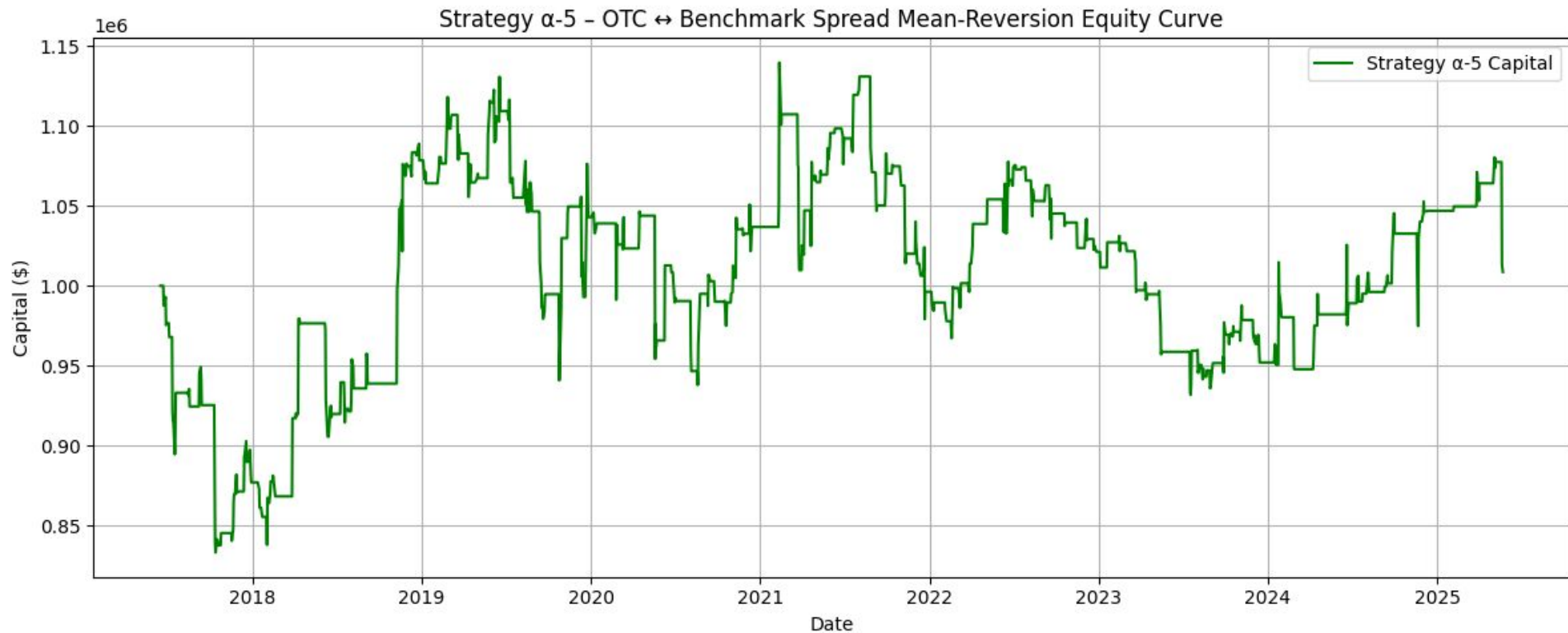
- **Data Used**

- OTC price (df_380) • Benchmark close series.

- **Objective**

- Capture temporary dislocations between OTC trades and posted MOPS benchmark; target Sharpe > 0.4 and max drawdown $< 25 \%$.

OTC $\leftrightarrow$ Benchmark Spread Mean-Reversion



OTC ↔ Benchmark Spread Mean-Reversion

- Total Return: 0.86%
- Sharpe Ratio: 0.08
- Max Drawdown: -18.22%
- Annual Volatility: 13.72%
- Win Rate: 11.73%
- Turnover: 0.04
- Avg Hold Period: 5.69 days
- # of Trades: 346
- CAGR: 0.11%